A Shape That Smooths Itself Away

A reader's guide to Perelman's proof of the Poincaré conjecture:
Ricci flow, surgery, and finite extinction

Zinan Xing

Exposition only — all mathematics is due to the cited authors.

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Abstract

In 1904 Poincaré asked whether a closed three-dimensional space in which every loop can be shrunk to a point must be the three-sphere. In 2002–2003 Grigori Perelman answered yes, completing Richard Hamilton's program of *Ricci flow*: let the geometry of the space evolve like heat, cut the thin necks where it pinches, and show that the pieces disappear in finite time. This guide is written for a reader who knows calculus and a little linear algebra, not differential geometry. It follows the shortest route to the Poincaré conjecture, which does not need the full geometrization theorem: the flow, the shapes that can form where curvature blows up, surgery, and the finite-extinction argument for simply connected spaces. Separately, it checks the statement of a recent Lean formalization clause by clause against Milnor's official problem text. Citations [X . . .] refer to source [X] by section, page or equation; [background: . . .] notes give context from the wider literature.

Where things stand

- **Proved**: every closed, simply connected 3-manifold is homeomorphic to the 3-sphere S^3. Perelman's three preprints [P1, P2, P3]; complete written accounts by Morgan–Tian [MT Intro., p. 5], Kleiner–Lott [KL] and Cao–Zhu [CZ]; an alternative final step by Colding–Minicozzi [CM].
- **Accepted**: the Clay Mathematics Institute announced the Millennium Prize for Perelman on March 18, 2010 [Clay p. 1]; he declined the prize money [AR p. 13], as he had declined the 2006 Fields Medal [IMU].
- **Formalized**: a Lean formalization of the full proof is public [L, LP]; Section 7 says what it certifies.
- **Still open nearby**: the *smooth* Poincaré conjecture in dimension four [Mil p. 5].

1 The question

Milnor's official problem description for the Clay Mathematics Institute states the question as Poincaré left it [Mil §1, p. 1]:

> *If a compact three-dimensional manifold M^3 has the property that every simple closed curve within the manifold can be deformed continuously to a point, does it follow that M^3 is homeomorphic to the sphere S^3?*

Here $S^3 = \{(x, y, z, w) \in \mathbb{R}^4 : x^2 + y^2 + z^2 + w^2 = 1\}$ [Mil p. 1]. The words, read one at a time:

- A **3-manifold** is a space that looks like ordinary $\mathbb{R}^3$ near every point, the way the surface of the Earth looks flat near every point.
- **Closed** means compact (finite in extent) and without boundary (no edge to fall off). Milnor writes "compact" in the question and "closed" in the next sentence [Mil p. 1].
- **Simply connected** means every loop can be shrunk to a point inside the space; formally, the fundamental group $\pi_1(M)$ is trivial [Mil fn. 4, p. 3].
- **Homeomorphic** means the same up to continuous stretching, with a continuous inverse.

The conjecture is the affirmative answer: *every simply connected closed 3-manifold is homeomorphic to the 3-sphere* [Mil p. 1].

Picture, not proof

One dimension down, the question is easy to picture. On a sphere, a rubber band can always be slid off to a point. On the surface of a doughnut, a band that goes around the hole cannot. Among closed surfaces, "every band slides off" singles out the sphere. Poincaré asked whether the same test singles out S^3 among closed 3-dimensional spaces, which we cannot see from the outside.

Why dimension three was the hard case. Poincaré first proposed a stronger claim, using homology instead of loops, and refuted it himself in 1904 with the icosahedral space, whose fundamental group is a perfect group of order 120 [Mil §2, p. 2]. In higher dimensions the analogue was settled first. For $n \geq 5$, Smale proved that a smooth homotopy n-sphere is homeomorphic to S^n [Mil §3, p. 4]; one reason high dimensions are easier is the room to move 2-dimensional disks apart [Mil p. 3]. For $n = 4$, Freedman's classification gives the topological statement [Mil pp. 4–5]. (In dimensions above three, "simply connected" is not enough: $S^2 \times S^2$ is simply connected but is not a sphere, so one asks for a homotopy sphere [Mil fn. 4, p. 3].) Dimension three resisted every topological attack, including a false proof by Whitehead in 1934 [Mil pp. 2–3].

Topological or smooth: the same question in dimension three. In dimension three, "the discrepancies between topological, piecewise linear, and differentiable theories disappear" [Mil p. 5]: every topological 3-manifold carries a smooth structure, and homeomorphisms can be approximated by diffeomorphisms (Moise, then Munkres [AR p. 10]; see also [MT Intro., fn. 1, p. 5]). So it is enough to prove the smooth version, which is what Ricci flow, a tool of calculus, can attack.

2 The theorem, in words and in symbols

Theorem (Perelman [P1, P2, P3]; as stated in [MT Cor. 0.2(a), p. 6])**.** *A closed, simply connected 3-manifold is diffeomorphic to* S^3.

By the remark above, the topological form follows: if M is a closed topological 3-manifold with $\pi_1(M) = 1$, give it a smooth structure (Moise–Munkres), apply the theorem, and forget smoothness.

In words: there is only one closed 3-dimensional space in which every loop can be shrunk to a point, and it is the round three-sphere.

What the theorem does *not* say. It does not classify all closed 3-manifolds. That is Thurston's geometrization conjecture, which Perelman's papers also address [P1 §13]; [P2 §§7–8] and whose proof was completed in detail later [MTG, BBBMP]. The route followed here never needs it: Perelman himself notes that for these manifolds "the analysis of the long time behavior … is not needed" [P3 Rem. 1.4, p. 3], and Morgan–Tian's account "avoids all references to the nature of the flow as time goes to infinity" [MT p. 8]. Nor does it say anything about dimension four in the smooth category, which is still open (Section 8).

3 The key idea: let the geometry flow like heat

A smooth manifold can be given a *Riemannian metric* g: a rule for measuring lengths and angles at every point. From g one computes curvature. In three dimensions the curvature that matters here is the *Ricci curvature* Ric, which at each point is a symmetric 3×3 matrix, like g itself, and its trace the *scalar curvature* R. Hamilton's *Ricci flow* lets the metric evolve in time by

$$\frac{\partial g}{\partial t} \;=\; -2\,\mathrm{Ric}(g) \tag{1}$$

[P1 p. 1]; [Mil §5, p. 6]. Where the space is positively curved, distances shrink; where it is negatively curved, they grow. Written in good coordinates, (1) is a heat equation for the metric, and the scalar curvature obeys $\partial_t R = \Delta R + 2|\,\mathrm{Ric}\,|^2$ [P1 p. 1]. Like heat, the flow tends to even things out; unlike heat, it has the extra term $2|\,\mathrm{Ric}\,|^2$, which can make curvature run away.

Picture, not proof

Think of a lumpy balloon whose rubber relaxes toward a uniform tension. Bumps smooth out. If the balloon is round everywhere, it simply shrinks. The hope of the program: start from *any* metric on M, let it flow, and watch it become round, which would reveal M as S^3.

A computation by hand: the shrinking sphere and the shrinking neck

For the round n-sphere of radius r, $\mathrm{Ric} = \frac{n-1}{r^2}\, g$. Write $g(t) = r(t)^2\, g_{\text{unit}}$; then (1) reads $\frac{d}{dt}(r^2) = -2(n-1)$, so

$$S^3: \quad r(t)^2 = r_0^2 - 4t, \qquad\qquad S^2 \times \mathbb{R} \text{ (a "neck")}: \quad r(t)^2 = r_0^2 - 2t,$$

since the line factor of the neck has zero curvature and does not move [MT §3.2, p. 64]. The 3-sphere vanishes at $T = r_0^2/4$; a neck pinches at $T = r_0^2/2$. The scalar curvature of the sphere is $R = 6/r^2$ [Mil p. 6], so it blows up as $t \to T$. A positive lower bound $R \geq a$ forces any solution to die by time $3/(2a)$ [P2 §6.1, p. 13]; for the round sphere, $a = 6/r_0^2$ gives exactly $r_0^2/4$.

Hamilton's first success, and the obstacle. In 1982 Hamilton proved that if the initial metric has positive Ricci curvature, the flow shrinks M to a point while becoming round, and after rescaling converges to a metric of constant positive curvature; so M is S^3 or a quotient of it [MT Intro. §1, pp. 6–7]; [H]. For a general starting metric this fails: curvature can blow up on a small region while the rest of the space stays tame. The model is a dumbbell, two round lobes joined by a thin neck. The neck has high positive curvature around its circumference and pinches off first (Figure 1, panels 1–2). The whole difficulty of the proof is to understand every such singularity and to continue past it.

4 What can form where curvature blows up

Perelman's first paper [P1] supplies the tools that let one zoom in on a singularity and recognize what is there. Four of them carry the argument.

1. Curvature becomes almost positive. The Hamilton–Ivey estimate says that in three dimensions, "at the points where scalar curvature is large, the negative part of the curvature tensor is small compared to the scalar curvature" [P1 p. 2]; precise form [MT Thm. 4.26, p. 100]. So any blow-up limit has nonnegative curvature.

2. An entropy that only increases. Perelman introduced a functional $\mathcal{W}(g, f, \tau)$ [P1 (3.1), p. 8], which he describes as "in a sense analogous to minus entropy" [P1 §5, p. 11], and proved that along the flow

$$\frac{d\mathcal{W}}{dt} \;=\; \int_M 2\tau \left| \mathrm{Ric} + \nabla\nabla f - \frac{g}{2\tau} \right|^2 (4\pi\tau)^{-n/2} e^{-f}\, dV \;\geq\; 0 \tag{2}$$

[P1 (3.4), p. 8]. A quantity that can only move one way rules out many bad scenarios, much as the second law of thermodynamics rules out perpetual motion. Equality holds only for special self-similar solutions (*gradient shrinking solitons*); the shrinking round sphere and the shrinking round cylinder are examples. A second monotone quantity, the *reduced volume* [P1 §7.1, p. 18], does similar work.

3. No local collapsing. The key consequence, which Perelman says "removes the major stumbling block in Hamilton's approach" [P1 p. 4]: a region cannot become thin without having large

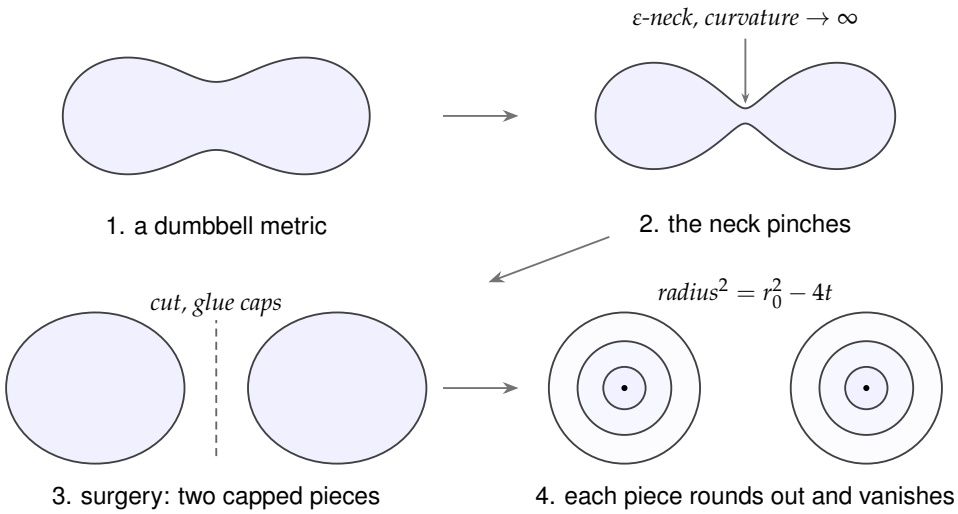


Figure 1: The story of the proof on a dumbbell. A schematic picture, not a computed flow: the real flow lives on a 3-manifold and its necks are $S^2\times$(interval), drawn here as their 2-dimensional shadows.

curvature. Precisely, the flow is κ-*noncollapsed*: every ball of radius r below a fixed scale on which $|\mathrm{Rm}| \le r^{-2}$ has volume at least κr^3 [P1 Def. 4.2, p. 10]; the theorem is [P1 Thm. 4.1, p. 10], with a local version [P1 Thm. 8.2, p. 20]. This is what makes blow-up limits exist as honest 3-dimensional flows, rather than collapsing to something of lower dimension.

4. The canonical neighborhoods. Putting these together, Perelman classifies the possible blow-up limits (*ancient κ-solutions* [P2 §1.1, p. 2]: flows defined for all negative times, nonnegatively curved and κ-noncollapsed [P1 §11.1, pp. 27–28]) and proves that near any point of high curvature, the flow, after rescaling, is close to one of them [P1 Thm. 12.1, p. 32]. Concretely, every such point has a neighborhood close to one of a few standard shapes [P2 §3, p. 6], the four kinds of neighborhood found in ancient κ-solutions [P2 §1.5, p. 4]:

shape	what it looks like	source
(a) ε-neck	a long piece of the round cylinder $S^2 \times \mathbb{R}$	[P2 p. 1, §1.5]
(b) ε-cap	a ball (or $\mathbb{R}P^3$ minus a ball) closing off a neck	[P2 p. 2, §1.5]
(c) closed	a whole component diffeomorphic to S^3 or $\mathbb{R}P^3$	[P2 §1.5, p. 4]
(d) closed, round	a whole component of constant positive curvature	[P2 §1.5, p. 4]

So where curvature is very large, the space is either a whole component of a known type (S^3, $\mathbb{R}P^3$ or a quotient of the round S^3), or it contains long thin tubes $S^2\times$(interval), possibly closed off by caps. The tubes are exactly the places to cut.

5 The main steps

5.1 Run the flow until the first singular time

Start from any metric on M. The flow exists for a short time and runs until curvature blows up somewhere. Perelman describes the picture at that first singular time T [P2 §3, pp. 6–7]: there is an open set Ω where the metric has a smooth limit. If Ω is empty, the whole manifold becomes singular at once, and M is S^3, $\mathbb{R}P^3$, $S^2 \times S^1$ or $\mathbb{R}P^3 \# \mathbb{R}P^3$. Otherwise the high-curvature parts are horns and tubes made of ε-necks, and M is a *connected sum* of the pieces of Ω closed off by caps, together with finitely many copies of $S^2 \times S^1$ and $\mathbb{R}P^3$ [P2 §3, p. 7].

Picture, not proof

A connected sum $A \# B$ is what you get by removing a small ball from each of A and B and gluing the two spherical holes together: two balloons joined by a tube. Cutting the tube and capping both ends undoes it. Gluing on a copy of S^3 changes nothing: $A \# S^3 = A$.

5.2 Surgery: cut the necks, cap the ends, continue

Deep inside each horn, Perelman finds a *δ-neck*, a stretch that is extremely close to a round cylinder of small radius h [P2 Lemma 4.3, p. 8]. Surgery then does what the picture suggests: "cut it along the middle two-sphere, remove the horn-shaped end, and glue in an almost standard cap" [P2 §4.4, pp. 8–9]. The cap is modeled on a specific rotationally symmetric model flow, the *standard solution* [P2 §2, pp. 4–6]. Components that are already close to round are recognized and declared extinct [P2 §4.4, p. 9]. The flow then restarts on the new manifold (Figure 1, panel 3).

Topologically, each surgery cuts along a 2-sphere and caps the two sides with balls: it undoes a connected sum. Each discarded component is diffeomorphic to $S^2 \times S^1$, to $\mathbb{R}P^3 \# \mathbb{R}P^3$, to the non-orientable S^2-bundle over S^1, or to a manifold with a metric of constant positive curvature [MT Thm. 0.3, p. 8].

5.3 Surgeries do not pile up

Two things could still go wrong: surgeries could accumulate at some finite time, and after a surgery the tools of Section 4 might stop working. Perelman rules out the first with a volume count: "every surgery reduces the volume by at least h^3, the sequence of surgery times is discrete" [P2 §4.4, p. 9]. On a finite time interval the volume stays bounded above (under the flow $dV/dt = -\int R\, dV$, and R is bounded below) and h stays bounded below, so only finitely many surgeries fit. The second is the heart of the second paper: the parameters of surgery can be chosen, decreasing with time, so that noncollapsing and the canonical neighborhoods survive every surgery, and the *Ricci flow with surgery* is "defined for $t \in [0, +\infty]$" [P2 Prop. 5.1, p. 11]. In Morgan–Tian this is Theorem 15.9 [MT pp. 353–354]. The point is to make the surgery scale h "arbitrarily small while keeping r bounded away from zero" [P2 p. 1], where r is the scale below which canonical neighborhoods are guaranteed.

5.4 For simply connected M, everything dies in finite time

So far the flow with surgery exists forever, and at each surgery the topology changes in a controlled way. For general M one would have to understand what happens as $t \to \infty$, which is

where geometrization lives. For the Poincaré conjecture there is a shortcut, proved in Perelman's third paper:

Theorem ([P3 Thm. 1.1, p. 1]). *Let M be a closed oriented three-manifold whose prime decomposition contains no aspherical factors. Then for any initial metric on M the solution to the Ricci flow with surgery becomes extinct in finite time.*

A simply connected closed 3-manifold satisfies the hypothesis: it is orientable, and an aspherical closed 3-manifold has infinite fundamental group, so it cannot be a factor of a manifold with $\pi_1 = 1$ [background: standard 3-manifold topology]. In fact Perelman remarks that for a homotopy sphere the argument can track a single homotopy class directly [P3 Rem. 1.4, p. 3].

The idea of the proof. Because M is not aspherical, some family of loops in M cannot be contracted through loops in a trivial way (a nontrivial class in a homotopy group of the loop space) [P3 p. 1]. Each loop in the family bounds disks; let $A(t)$ be the min-max area: the least, over such families, of the largest least-area disk in the family [P3 p. 1]. Perelman shows

$$\frac{dA}{dt} \;\le\; -2\pi - \tfrac{1}{2} R_{\min}(t)\, A \tag{3}$$

[P3 Lemma 1.2, p. 2], using minimal disks, the Gauss–Bonnet formula and a curve-shortening flow. The curvature itself is bounded below, $R_{\min}(t) \ge -\frac{3}{2(t+c)}$, by the maximum principle [P3 §1.3, p. 3]. Surgeries do not spoil the estimate: they distort lengths by at most a factor close to one [P3 §1.3, pp. 2–3].

Why (3) forces extinction

With $R_{\min} \ge -\frac{3}{2(t+c)}$, (3) gives $\frac{dA}{dt} \le -2\pi + \frac{3A}{4(t+c)}$. Put $\hat{A} = A/(t+c)^{3/4}$; then $\frac{d\hat{A}}{dt} \le -2\pi\,(t+c)^{-3/4}$, so

$$\hat{A}(t) \;\le\; \hat{A}(0) - 8\pi\big((t+c)^{1/4} - c^{1/4}\big) \;\longrightarrow\; -\infty.$$

An area cannot be negative, so the flow cannot exist past a finite time: it becomes extinct. (Perelman [P3 §1.3, p. 3] rescales by $A/(t+c)$ instead, which gives a logarithm in place of the fourth root; the conclusion is the same.)

Colding and Minicozzi gave a second proof of this step with a different min-max quantity, the *width*: sweep M out by a family of 2-spheres and take the least possible largest area. They prove $\frac{d}{dt}W \le -4\pi + \frac{3}{4(t+C)}W$, "hence, $g(t)$ must become extinct in finite time" [CM Thm. 1.1, p. 2]; a version with complete proofs "intended for a general non-specialist audience" is [CM2]. In both arguments the mechanism is the same: the flow squeezes a min-max area at a definite rate, faster than the area can ever be replenished.

6 Finishing the proof

Now run the story backward from the extinction time T. When the flow with surgery dies, every remaining piece was a standard one, and each surgery undid a connected sum along a 2-sphere. Reassembling:

$$M \;\cong\; (\text{spherical space forms } S^3/\Gamma) \,\#\, \cdots \,\#\, (\text{copies of } S^2 \times S^1)$$

[MT Thm. 0.1, p. 5; Cor. 0.5, pp. 8–9]. Now use that $\pi_1(M) = 1$. The fundamental group of a connected sum is the free product of the fundamental groups of the pieces (van Kampen's theorem), so every piece is simply connected. That rules out $S^2 \times S^1$ ($\pi_1 = \mathbb{Z}$) and every quotient S^3/Γ with $\Gamma \neq 1$. What remains is a connected sum of copies of S^3, and $S^3 \# S^3 \# \cdots = S^3$. Hence

$$M \cong S^3.$$

This is the outline given by Kleiner and Lott [KL §3.2, pp. 8–9]. Finally, by the Moise–Munkres theorem, the smooth conclusion gives the topological one that Clay asked for (Section 1).

How the proof was checked by people. Perelman posted his three papers to arXiv in 2002–2003 and did not take "the usual route of submission of his work to a refereed journal" [AR p. 2]. The details were supplied by several groups. Kleiner and Lott found "some incorrect statements and incomplete arguments" but "did not find any serious problems, meaning problems that cannot be corrected using the methods introduced by Perelman" [KL §1, p. 1]. Morgan and Tian wrote out the whole argument for the Poincaré conjecture "in great detail", partly because of "the number of incorrect claims of proofs of these results in the past" [MT p. 6]. The Clay Institute's decision passed through a special advisory committee, its scientific advisory board and its board of directors [Clay p. 1].

7 What the Lean formalization certifies

In 2026 Qin, Liao, Khaitan and Chow released a formalization of the Hamilton–Perelman proof in the Lean 4 proof assistant, built on the Mathlib library [L, LP]. A proof assistant does not trust the authors: a small program, the *kernel*, rechecks every logical step down to the axioms.

7.1 How the trust is arranged

What must be trusted is small: Lean's kernel, Mathlib's definitions of the words in the statement, and the statement itself. The project is several million lines long, but the statement uses only standard Mathlib notions, so none of the authors' own definitions can change what is being claimed; they can only affect whether the proof checks. The authors report that the theorem depends only on Lean's three standard axioms (propositional extensionality, the axiom of choice, and soundness of quotients), with no unproved step [L]. Hamilton's and Perelman's results, such as the pinching estimate, the entropy and reduced-volume monotonicity, noncollapsing, canonical neighborhoods and finite extinction with surgery, are proved inside the project rather than assumed, and so is the existence of a smooth structure on a topological 3-manifold [L]. MathDB, the platform hosting the problem, records a submission of this proof (with a short bridge to its own target statement, below) as "Lean verified" after a kernel replay and an axiom audit [MDB].

7.2 Statement audit: the Lean text against the official text

The main theorem reads [L]:

```
theorem poincare_conjecture
    (M : Type u) [TopologicalSpace M]
    [ChartedSpace (EuclideanSpace ℝ (Fin 3)) M]
    [T2Space M] [CompactSpace M] [SimplyConnectedSpace M] :
    Nonempty (M ≃ₜ Metric.sphere (0 : EuclideanSpace ℝ (Fin 4)) 1)
```

Clause by clause against Milnor's text [Mil p. 1]:

Official clause	Lean	Reading
3-dimensional manifold	`ChartedSpace (EuclideanSpace ℝ (Fin 3)) M`	an atlas of charts onto open sets of $\mathbb{R}^3$
(Hausdorff, part of "manifold")	`T2Space M`	stated explicitly
closed: compact	`CompactSpace M`	as stated
closed: no boundary	charts modeled on $\mathbb{R}^3$, not a half-space	no boundary points
(connected)	implied by `SimplyConnectedSpace M`	as stated
simply connected	`SimplyConnectedSpace M`	trivial fundamental groupoid, i.e. $\pi_1 = 1$
the sphere S^3	`Metric.sphere (0 : EuclideanSpace ℝ (Fin 4)) 1`	$\{x \in \mathbb{R}^4 : \|x\| = 1\}$
homeomorphic	`Nonempty (M ≃ₜ ...)`	a homeomorphism exists

We found no discrepancy. The Lean statement is the topological form posed by Clay, with no extra hypothesis (no smoothness, metric or orientation is assumed; these are constructed inside the proof). It allows M in any universe of types, which only makes it more general. A smooth version is proved as well [L]. This is a reading-level check by the author, not reviewed by a Lean expert.

MathDB's target statement. MathDB's own formal target, adapted from Google DeepMind's Formal Conjectures, is phrased differently: a Hausdorff 3-manifold that is *homotopy equivalent* to S^3 is homeomorphic to S^3 [MDB]. The submission bridges the two by deriving compactness and simple connectivity from the homotopy equivalence and then applying the theorem above [MDB]. Both forms are correct statements of the conjecture; the one displayed above is the closer match to Milnor's text.

7.3 What Lean does not certify

- That Mathlib's definitions mean what a topologist means. A person has to read them; the table above is that reading.
- That the Lean proof follows Perelman's papers step by step. The kernel checks correctness, not faithfulness to a particular text.
- Anything in this essay. The prose here is an informal guide and may simplify.

- The trusted base itself: the kernel and the toolchain are assumed correct, as in any formal proof.

8 What remains open

- **The smooth 4-dimensional Poincaré conjecture**: is every smooth homotopy 4-sphere diffeomorphic to S^4? Milnor lists it as unknown [Mil p. 5]; it is still open [background: no proof or counterexample is known]. Ricci flow in dimension four meets singularities that are not yet understood.
- **Nothing in dimension three.** The Poincaré conjecture and, with further work, Thurston's geometrization conjecture are theorems [P2, MTG, BBBMP].

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References

[P1] G. Perelman, *The entropy formula for the Ricci flow and its geometric applications*, arXiv:math/0211159, 2002, https://arxiv.org/abs/math/0211159.

[P2] G. Perelman, *Ricci flow with surgery on three-manifolds*, arXiv:math/0303109, 2003, https://arxiv.org/abs/math/0303109.

[P3] G. Perelman, *Finite extinction time for the solutions to the Ricci flow on certain three-manifolds*, arXiv:math/0307245, 2003, https://arxiv.org/abs/math/0307245.

[Mil] J. Milnor, *The Poincaré Conjecture*, Clay Mathematics Institute official problem description, 2000, revised 2004, https://www.claymath.org/wp-content/uploads/2022/06/poincare.pdf.

[MT] J. Morgan and G. Tian, *Ricci Flow and the Poincaré Conjecture*, Clay Mathematics Monographs 3, AMS, 2007; arXiv:math/0607607, https://arxiv.org/abs/math/0607607.

[MTG] J. Morgan and G. Tian, *Completion of the proof of the geometrization conjecture*, arXiv:0809.4040, 2008, https://arxiv.org/abs/0809.4040.

[KL] B. Kleiner and J. Lott, *Notes on Perelman's papers*, Geom. Topol. 12 (2008); arXiv:math/0605667, https://arxiv.org/abs/math/0605667.

[CZ] H.-D. Cao and X.-P. Zhu, *Hamilton–Perelman's proof of the Poincaré conjecture and the geometrization conjecture*, arXiv:math/0612069, 2006, https://arxiv.org/abs/math/0612069.

[CM] T. H. Colding and W. P. Minicozzi II, *Estimates for the extinction time for the Ricci flow on certain 3-manifolds and a question of Perelman*, arXiv:math/0308090, https://arxiv.org/abs/math/0308090.

[CM2] T. H. Colding and W. P. Minicozzi II, *Width and finite extinction time of Ricci flow*, arXiv:0707.0108, https://arxiv.org/abs/0707.0108.

[BBBMP] L. Bessières, G. Besson, M. Boileau, S. Maillot and J. Porti, *Geometrisation of 3-Manifolds*, EMS Tracts in Mathematics 13, EMS, 2010.

[H] R. S. Hamilton, *Three-manifolds with positive Ricci curvature*, J. Differential Geom. 17 (1982), 255–306.

[Clay] Clay Mathematics Institute, *Prize for resolution of the Poincaré conjecture awarded to Dr. Grigoriy Perelman*, press release, March 18, 2010, https://www.claymath.org/wp-content/uploads/2022/06/Poincare-press-release.pdf.

[AR] Clay Mathematics Institute, *Annual Report 2010*, https://www.claymath.org/library/annual_report/ar2010/ar2010.pdf.

[IMU] International Mathematical Union, *Fields Medals 2006*, https://www.mathunion.org/imu-awards/fields-medal/fields-medals-2006.

[L] Z. Qin, Y. Liao, A. Khaitan and B. Chow, *differential-geometry* (Lean 4 formalization), commit `f14e302a`, https://github.com/qinz1yang/differential-geometry; theorem `DifferentialGeometry.Topology.poincare_conjecture`.

[LP] Z. Qin, Y. Liao, A. Khaitan and B. Chow, *A Lean formalization of the Hamilton–Perelman proof of the three-dimensional Poincaré conjecture*, arXiv:2609.33842, 2026.

[MDB] MathDB, *Poincaré conjecture*, problem post 316224 (formal target and verification record), https://mathdb.com/p/316224/poincare-conjecture; submission fork commit `9f65eaf9`, https://github.com/liao9yuan/differential-geometry-mathdb.